

TOPS TECH Task 12 (CS)

Task-11 Theory (1 hour): Introduction to Virtualization What is virtualization, types, and benefits? Overview of hypervisors: Type 1 (bare-metal) and Type 2 (hosted).

Practical (2 hours): Install and configure a hypervisor (VirtualBox or VMware Workstation). Create a virtual machine (VM).

- 1. Install VirtualBox or VMware Workstation on your computer and take a screenshot of the installation screen showing the software is installed.**

Ans:

Installation of VMware Workstation

Objective:

- To install VMware Workstation on a Windows computer and verify that the virtualization software is functioning correctly.

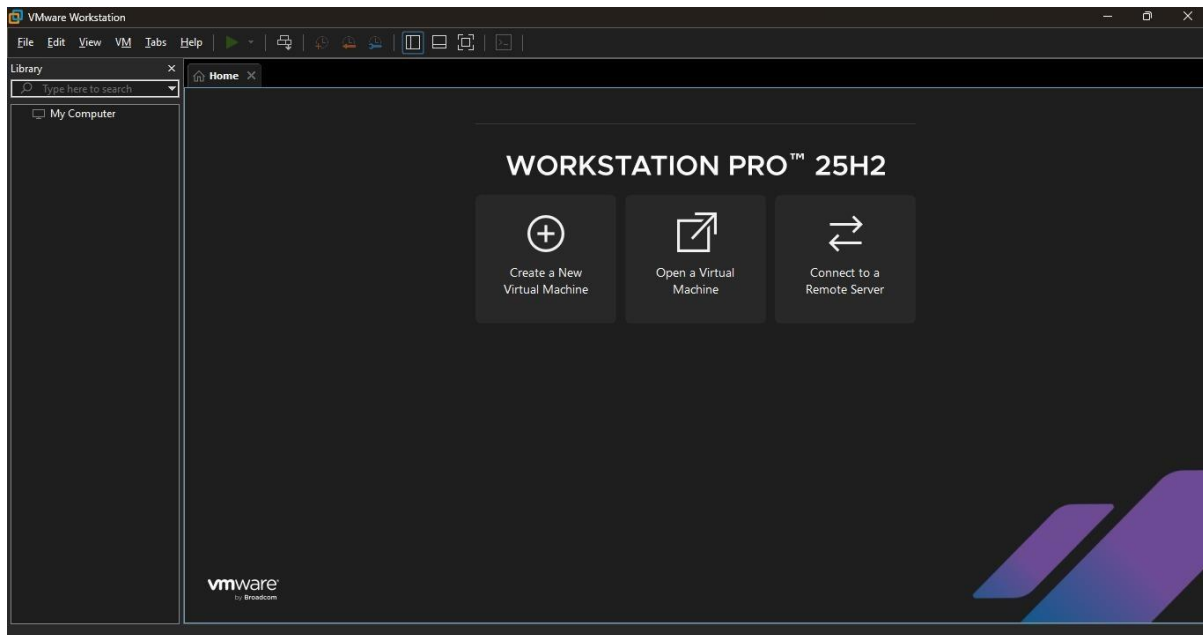
Steps Performed:

- Downloaded VMware Workstation from the official VMware website.
- Executed the installer file with administrative privileges.
- Followed the installation wizard and accepted the license agreement.
- Selected the default installation options.
- Completed the installation process successfully.
- Restarted the computer if required.
- Launched VMware Workstation from the Start Menu.
- Verified that the VMware Workstation interface opened successfully.

Result:

- VMware Workstation was successfully installed on the system and is ready for creating and running virtual machines.

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2. **Create a new virtual machine in VirtualBox or VMware Workstation and configure it to run Ubuntu Linux as the guest OS with at least 2GB RAM and 20GB storage.**
Hint: Download the Ubuntu ISO file from the official website if you don't already have it.

Ans:

Creation and Configuration of Ubuntu Virtual Machine

Objective:

- To create and configure a virtual machine using VMware and install Ubuntu Linux as the guest operating system.

Steps Performed:

- Opened VMware.
- Clicked on "Create a New Virtual Machine".
- Selected the Typical (Recommended) configuration option.
- Chose the Ubuntu ISO image file as the installation media.
- Selected Linux as the guest operating system.

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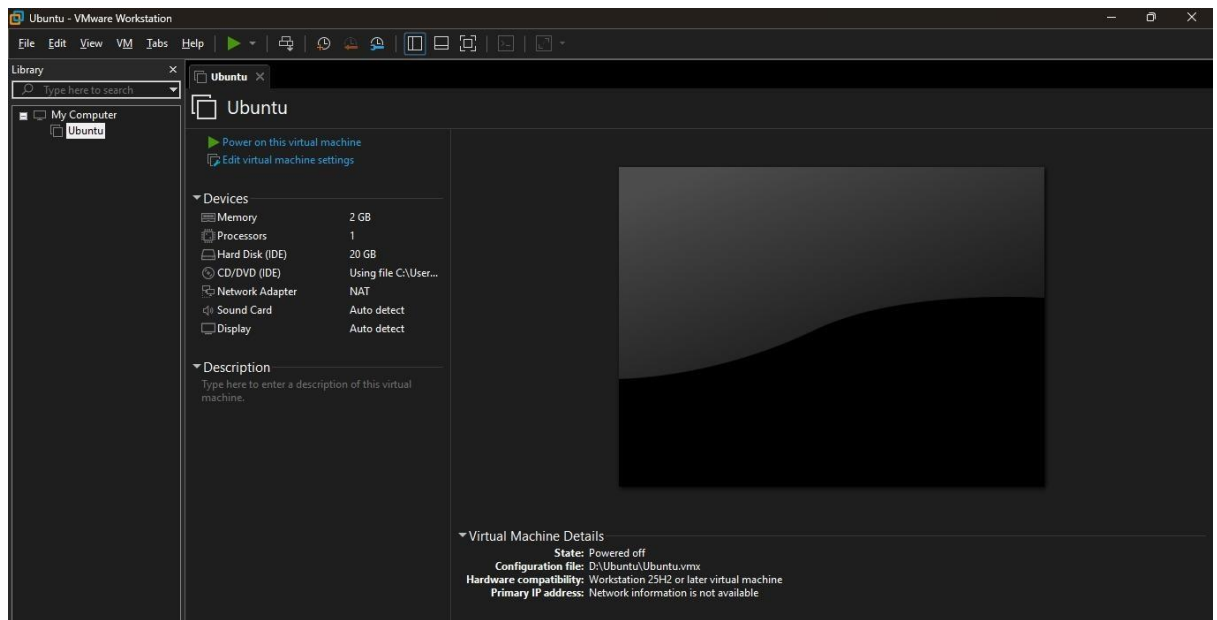
- Selected Ubuntu 64-bit as the operating system version.
- Assigned the virtual machine name as "Ubuntu".
- Allocated 2 GB (2048 MB) RAM to the virtual machine.
- Allocated 20 GB virtual disk storage.
- Completed the virtual machine creation process.
- Powered on the virtual machine and started the Ubuntu installation.
- Followed the Ubuntu installation wizard and completed the installation successfully.

Virtual Machine Configuration:

- Virtual Machine Name: Ubuntu
- Operating System: Ubuntu Linux
- RAM Allocated: 2048 MB (2 GB)
- Virtual Hard Disk Size: 20 GB
- Hypervisor Used: VMware

Result:

- The Ubuntu virtual machine was successfully created and configured in VMware with 2 GB RAM and 20 GB storage.
- The system is ready for use as a guest operating system.



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- 3. Start your Ubuntu virtual machine and take a screenshot of the Ubuntu desktop running inside the VM window.**

Ans:

Booting and Verifying Ubuntu Virtual Machine

Objective:

- To start the Ubuntu virtual machine in VMware and verify the successful installation and operation of the Ubuntu operating system.

Steps Performed:

- Opened VMware.
- Selected the previously created Ubuntu virtual machine.
- Clicked on the "Power On" button to start the virtual machine.
- Waited for the Ubuntu operating system to boot successfully.
- Logged in using the configured username and password.
- Verified that the Ubuntu desktop environment loaded correctly.
- Confirmed that the operating system was functioning properly.

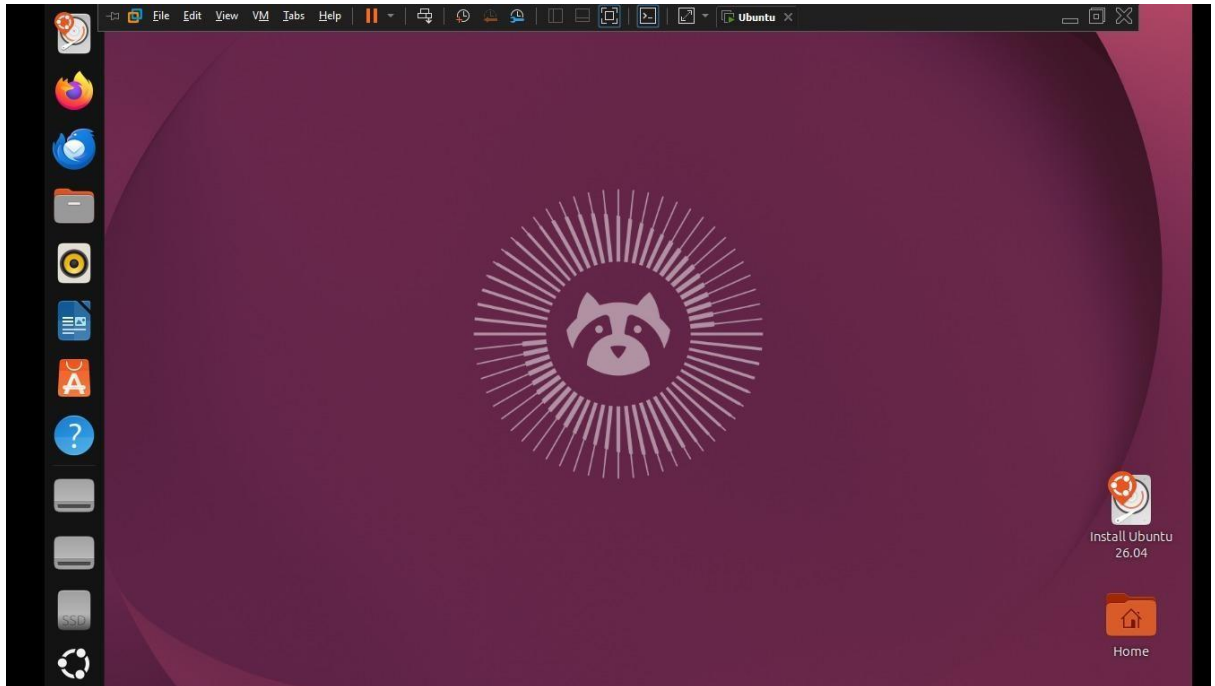
Verification:

- The Ubuntu operating system booted successfully without any errors.
- The desktop environment loaded correctly, and all basic system functions were operational.

Result:

- The Ubuntu virtual machine was successfully started in VMware.
- Ubuntu Linux booted successfully and the desktop environment was accessible and functioning properly.

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4. List 3 differences between Type 1 (bare-metal) and Type 2 (hosted) hypervisors, and name one real-world example of each.

Ans:

Comparison Between Type 1 and Type 2 Hypervisors

Objective:

- To study and compare the characteristics, architecture, performance, and use cases of Type 1 and Type 2 Hypervisors.

Introduction:

- A hypervisor is software that enables multiple virtual machines to run on a single physical computer. It manages hardware resources and allocates them to virtual machines. Hypervisors are classified into two categories: Type 1 Hypervisors and Type 2 Hypervisors.

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Comparison Between Type 1 and Type 2 Hypervisors:

<u>Feature</u>	<u>Type 1 Hypervisor</u>	<u>Type 2 Hypervisor</u>
Installation	Installed directly on physical hardware	Installed on top of a host operating system
Performance	Higher performance	Slightly lower performance
Security	More secure	Less secure compared to Type 1
Resource Usage	Uses hardware resources directly	Depends on host operating system resources
Management	Suitable for enterprise environments	Suitable for personal and educational use
Operating System Dependency	Does not require a host operating system	Requires a host operating system
Examples	VMware ESXi, Microsoft Hyper-V Server	VMware, Oracle VirtualBox

Advantages of Type 1 Hypervisor:

- Better performance.
- Improved security.
- Efficient resource utilization.
- Suitable for data centers and cloud environments.

Advantages of Type 2 Hypervisor:

- Easy to install and configure.
- Suitable for testing and learning purposes.
- Can run on existing operating systems.
- Ideal for students and personal use.

- ⇒ Type 1 Hypervisors are mainly used in enterprise environments because of their performance and security advantages.
- ⇒ Type 2 Hypervisors are commonly used for learning, development, and testing because they are easy to install and operate.

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- ⇒ VMware used in this practical is a Type 2 Hypervisor because it runs on top of the Windows operating system.
- ⇒ The differences between Type 1 and Type 2 Hypervisors were studied and understood successfully.